8.2 Proving Triangle Similarity by AA

Name _____ Period ____ Date _____

Geometry 8.2 · Student sheet

I can…

- use two pairs of congruent angles to show two triangles are similar.
- write a similarity statement with the vertices in the correct order.
- use similar triangles to find a missing length, including a height I can't measure directly.

Vocabulary · fill in the missing word

AA Similarity Theorem — if two angles of one triangle are _____ to two angles of another triangle, the triangles are similar

Indirect measurement — finding a length with similar _____ instead of measuring it

Reflexive Property — an angle is _____ to itself (a shared angle)

Vertical angles — opposite angles formed by two _____ lines; they are congruent

Part 1 · Discovery: two angles are enough

Draw a base of the length your teacher gives you: _____ cm. At one end draw a **40°** angle and at the other end a **65°** angle. Extend the sides until they meet.

	Base	Side next to 40°	Side next to 65°	Third angle
My triangle				
Partner's triangle				
Partner ÷ mine				—

What do you notice about the third angles? About the three ratios?

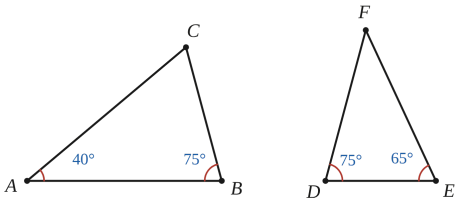
Part 2 · Notes and examples (fill in each note, then work the example)

AA Similarity Theorem: if two _____ of one triangle are congruent to two _____ of another triangle, then the triangles are _____.

Similarity statements: Write the similarity statement so the _____ angles are in the same positions.

Example 1

Determine whether the triangles are similar. If they are, write a similarity statement.



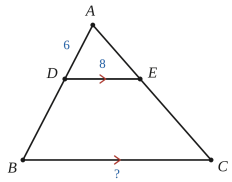
Example 2

△JKL has angles of 48° and 85°. △MNP has angles of 48° and 57°. Are the triangles similar? Explain.

Where congruent angles come from: Congruent angles can come from a shared angle, _____ angles, or _____ lines cut by a transversal.

Example 3

In △ABC, $\overline{DE} \parallel \overline{BC}$. Show that △ADE ~ △ABC. Then find BC when AD = 6, AB = 15, and DE = 8.



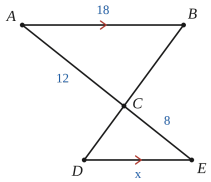
Indirect measurement: use _____ triangles to find a length you cannot measure directly (like a height from shadows).

Example 4

A 6-foot-tall person casts a 4-foot shadow. At the same time, a flagpole casts a 30-foot shadow. How tall is the flagpole?

Example 5

$\overline{AB} \parallel \overline{DE}$, and $\overline{AE}$ and $\overline{BD}$ intersect at C. AB = 18, AC = 12, and EC = 8. Find DE.



Part 3 · Partner practice

Partner 1

△PQR has m∠P = 90° and m∠Q = 35°. △STU has m∠S = 55° and m∠T = 90°. Are the triangles similar? If so, write a similarity statement.

Partner 2

A 5-foot-tall student casts a 3-foot shadow. At the same time, a tree casts a 24-foot shadow. How tall is the tree?